

Gateway Security Architecture

Production-grade security middleware for AI/ML inference

About

A security gateway between users and GPU-hosted AI models, blocking malicious files, credential abuse, replay attacks, and injection. Handles up to 50k-image batch uploads.

Tech Stack

Python 3.14 | Flask (4 microservices) | Pillow (image analysis) | hashlib/HMAC-SHA256
ThreadPoolExecutor (50 workers) | In-memory storage (no Redis) | SSH tunnel

Architecture Flow

Client --> Dashboard (:8080) --> Policy (:5002) --> Scanner (:5003) --> GPU (:5001)

Services

1. Mock GPU (:5001) Simulated AI inference (100-200ms/image)
2. Policy (:5002) API key auth, rate limits, concurrency, dedup, HMAC signing
3. Scanner (:5003) 8-format detection, magic bytes, image bombs, entropy flagging
4. Dashboard (:8080) UI, bulk upload, async batch pipeline (250/chunk)

Rate Limits

Interactive: 20/min per key, 100/min per IP | Batch: 100/min (separate pool)

Concurrency: 50 parallel per tenant (30s auto-clean)

Batch Pipeline (50k images)

<50 files -> sync per-file parallel | >=50 files -> async job

1. Save files to disk session dir
2. Process 250-file chunks via batch-authorize + batch-scan + GPU
3. Poll /job-status (progress) and /job-result (final output)

Security Coverage

Magic bytes, dimensions <=10000px, <=50MP, aspect ratio <=100:1

Compression ratio <=500:1, color depth <=16-bit, EXIF detection

Shannon entropy flagging (info only, never blocks high-detail photos)

Key expiry/inactive, per-IP/key rate limits, file hash dedup (1h TTL)

Optional HMAC-SHA256 request signing with replay protection

409 Conflict on duplicate files (same SHA-256 within 1 hour)

Deployment

Standalone Flask processes via Start-Process (no Docker)

Tunnel: ssh -R 80:localhost:8080 nokey@localhost.run